



Original Article

Higher vs standard mean arterial pressure target in the immediate postoperative period of liver transplantation to prevent acute kidney injury: A randomized clinical trial (LIVER-PAM)



Rodolpho Augusto de Moura Pedro^{1,*} , Bruna Carla Scharanch¹ ,
 Lucas de Oliveira Araújo¹ , Paula Sepulveda Mesquita¹ ,
 Luciana Severo Brandão¹ , Lúcia da Conceição Andrade² ,
 Wellington Andraus³ , Luís Augusto Carneiro D'Albuquerque^{1,3} ,
 Jean-Michel Constantin⁴ , Alexandre Biasi Cavalcanti^{5,6} ,
 Alberto Queiroz Faria⁷ , Luiz Marcelo Sá Malbouisson^{1,3,5}

¹ Liver and Gastroenterology Intensive Care Unit, Department of Gastroenterology, Faculdade de Medicina, Universidade de São Paulo, São Paulo, Brazil

² Nephrology Division, Hospital das Clínicas, Universidade de São Paulo, São Paulo, Brazil

³ Liver and Gastrointestinal Transplant Division, Department of Gastroenterology, Faculdade de Medicina, Universidade de São Paulo, São Paulo, Brazil

⁴ Department of Anesthesiology and Critical Care, Pitie-Salpetriere Hospital, Sorbonne University, GRC 29, AP-HP, DMU DREAM, Paris, France

⁵ Division of Anesthesiology, Hospital das Clínicas, Universidade de São Paulo, São Paulo, Brazil

⁶ HCor Research Institute, São Paulo, Brazil

⁷ Department of Gastroenterology, Universidade de São Paulo, São Paulo, Brazil

ARTICLE INFO

Keywords:

liver transplant

acute kidney injury

mean arterial pressure

MAKE28

ABSTRACT

Acute kidney injury (AKI) is a frequent complication after liver transplantation, with an incidence up to 81%, and is associated with impaired long-term renal function, graft survival, and increased mortality. Hypotension is a modifiable risk factor, and while maintaining a mean arterial pressure (MAP) >65 mmHg is standard practice, higher targets may improve outcomes. The LIVER-PAM trial was an open-label, single-center, randomized clinical trial including 174 adult patients treated in an intensive care unit after liver transplantation. Patients were randomized to a higher MAP target (85-90 mmHg) or a standard target (65-70 mmHg) during the first 24 postoperative hours. The primary

Abbreviations: AKI, acute kidney injury; CH, chronic hypertension; CI, confidence interval; EAD, early allograft dysfunction; HAT, hepatic artery thrombosis; ICU, intensive care unit; MAKE28, major adverse kidney events at 28 days; PNF, primary nongraft dysfunction; RR, relative risk; RRT, renal replacement therapy; WIT, warm ischemia time.

* Corresponding author. Av. Dr. Enéas Carvalho de Aguiar, 255, São Paulo, 05403-000, Brazil

E-mail address: rodolphopedro@hotmail.com (R.A. de Moura Pedro).

<https://doi.org/10.1016/j.ajt.2025.07.2482>

Received 15 April 2025; Received in revised form 27 June 2025; Accepted 25 July 2025

Available online 5 August 2025

1600-6135/© 2025 American Society of Transplantation & American Society of Transplant Surgeons. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

outcome was AKI incidence within 7 days. Secondary outcomes included major adverse kidney events at 28 days, graft rejection, and 11 additional endpoints. AKI occurred in 69.7% of patients in the higher MAP group and 69.4% in the standard MAP group (relative risk [RR], 1.00; 95% confidence interval (CI), 0.82–1.22; $P = .97$). Major adverse kidney events at 28 days was lower in the higher MAP group (56.1% vs 72.9%; RR 0.77, 95% CI, 0.61–0.96, $P = .02$). Graft rejection requiring pulse therapy was more frequent in the higher MAP group (12.5% vs 3.5%; RR, 3.54; 95% CI, 1.02–12.25; $P = .03$). A higher MAP target did not reduce AKI incidence within 7 days postliver transplantation.

1. Introduction

Acute kidney injury (AKI) is among the most common complications following liver transplantation, with reported incidence rates ranging from 15% to 81%.¹ AKI negatively impacts long-term renal function, graft survival, and overall mortality.² Its high incidence has multifactorial causes,³ with hypotension identified as a potentially modifiable risk factor.⁴

In surgical patients, hypotension has been associated with AKI in a dose-dependent manner,⁵ given the close relationship between renal perfusion and mean arterial pressure (MAP).⁶ Although a MAP threshold of ≥ 65 mmHg is conventionally considered safe,⁷ higher MAP levels could result in better renal outcomes in select critically ill populations,^{8,9} including liver transplant recipients, according to observational data.^{10,11}

In light of the limited availability of donor organs, identifying strategies to optimize postoperative outcomes is of paramount importance. However, existing literature on hemodynamic optimization following liver transplantation remains scarce and largely retrospective. To address this gap, we conducted a prospective, randomized clinical trial to test the hypothesis that targeting a higher MAP (85–90 mmHg) during the first 24 postoperative hours would reduce the incidence of AKI within 7 days, compared with standard care (MAP, 65–70 mmHg).

2. Methods

2.1. Study design

The LIVER-PAM trial is an open-label, randomized clinical study conducted in the intensive care unit (ICU) of the Gastroenterology Department at Hospital das Clínicas, Faculdade de Medicina da Universidade de São Paulo—a 14-bed specialized ICU within a public academic hospital in Brazil, a center with a high incidence of AKI following liver transplantation (75%) according to its ICU database.

The trial was conducted in accordance with the Declaration of Helsinki, approved by the Research Ethics Committee (Comissão de Ética para Análise de Projetos de Pesquisa [Research Ethics Committee] approval number 4 887 829), and registered at [ClinicalTrials.gov](https://clinicaltrials.gov) (NCT05068713). The study protocol has been previously published,¹² and the trial is reported in accordance with the Consolidated Standards of Reporting Trials guidelines for randomized clinical trials¹³ (supplementary figure

1). All authors had full access to the study data and reviewed and approved the final manuscript.

2.2. Participants

All adult patients (≥ 18 years old) admitted to the ICU following liver transplantation were screened for eligibility. Informed consent was obtained from all participants or their legally authorized representatives.

Exclusion criteria included combined liver-kidney transplantation; treatment for hepatorenal syndrome within 48 hours before surgery or renal replacement therapy (RRT) in the 15 days before transplantation; transplantation for acute liver failure; persistent refractory shock during surgery or upon ICU admission (noradrenaline requirement >1 $\mu\text{kg}/\text{min}$ with a second vasopressor for >2 hours); cardiac arrest during transplantation or at ICU admission; contraindication to prolonged vasopressor use; treating physician's refusal to enroll; pregnancy; and liver retransplantation within 6 months of a previous procedure.

2.3. Randomization and masking

Randomization was required within 2 hours of ICU admission. A simple 1:1 allocation sequence was generated online (<https://www.randomizer.org>) and uploaded into a centralized electronic system (RedCap) by a research member who was not involved in eligibility screening, consent procedures, or clinical care. Randomized patients were followed for 28 days or until death, whichever occurred first.

Blinding the MAP values was not feasible. To minimize bias, data collection was performed by personnel not involved in clinical management and was overseen by the principal investigator. Statistical analysis was conducted by a statistician blinded to MAP values and group assignments. Given the safety profile reported in prior studies,^{8,9,14} no interim analyses were performed.

2.4. Procedures

Both groups received standard care, which included maintaining oxygen saturation $\geq 94\%$, serum hemoglobin levels ≥ 7 g/dL, body temperature $\geq 35^\circ\text{C}$, and cardiac index ≥ 2.5 L/min/m². MAP was continuously monitored during the intervention using

an indwelling arterial catheter. Standard hemodynamic monitoring also included the use of a pulmonary artery catheter and ultrasonography. Fluid infusion rate and type were left to the discretion of the treating clinicians, based on the patient's clinical history and physical examination.

In the intervention group, the MAP target value was set between 85 and 90 mmHg during the first 24 postoperative hours following ICU admission, whereas the control group had a MAP target of 65 to 70 mmHg during the same period. This higher target was chosen based on previous data¹⁴ and the anticipation that a minor final difference would remain in the final analysis,⁸ given that patients who spontaneously achieved higher MAP values after complete vasopressor weaning did not receive antihypertensive therapy to reduce MAP into the target range due to safety concerns. Although fluid responsiveness assessment was recommended to guide fluid therapy, the attending physician retained discretion to individualize the strategy using fluids and/or vasopressors when MAP levels were below the target. Norepinephrine was the first-line vasopressor, with vasopressin added as needed. After the intervention period, the MAP target was standardized to 65 to 70 mmHg for both groups until ICU discharge.

Antibiotic prophylaxis was defined before randomization, and immunosuppressive therapy was managed by the surgical team according to the institutional protocol (see [Supplementary topic 10](#)). Pulse therapy was prescribed on a case-by-case basis at the discretion of the same team.

2.5. Outcomes

The primary outcome was the incidence of AKI within 7 days following liver transplantation. As previously published,¹² AKI was defined as an absolute increase in serum creatinine levels to at least 1.5 times the baseline value, in accordance with the definition used in the previous cohort from the same institution, which served as the basis for the sample size calculation. Patients who died within 7 days of surgery were considered positive for the primary outcome.

Secondary outcomes were grouped into 4 sections:

1. Renal outcomes include major adverse kidney events at 28 days (MAKE28), defined as a composite outcome including death, dependence on RRT, or a persistent decrease in glomerular filtration rate of $\geq 25\%$ by days 28^{15–17}; RRT within 7 days; days alive and free of AKI in 7 days; and AKI measured using urinary neutrophil gelatinase-associated lipocalin levels at 48 hours.
2. Hepatic outcomes include retransplantation in 28 days, early allograft dysfunction (EAD), primary nongraft dysfunction (PNF), hepatic artery thrombosis (HAT) in 28 days, and graft rejection requiring pulse therapy in 7 days.
3. General outcomes include survival at 28 days, days alive and free of ICU in 7 days, days alive and free of hospital in 28 days, and surgical site infection in 7 days.
4. Safety outcomes include major bleeding in 7 days, blood product consumption in 7 days, and arrhythmia requiring electrical or chemical treatment in 7 days.

Specific outcome definitions have been previously published¹² and are available in the [Supplementary topic 1](#).

2.6. Statistical analysis

Based on a previously observed AKI incidence of approximately 75% in a cohort from the same institution, a sample size of 174 patients was calculated to detect an absolute reduction of 20% in AKI incidence. This calculation assumed a statistical power of 80% and a 2-sided alpha level of 5%.

Statistical analysis was performed according to the intention-to-treat principle. Descriptive statistics are presented as frequency percentage for categorical variables, median (interquartile range, 25%-75%) for continuous nonparametric values, and mean ($\pm$ standard deviation) for parametric distribution. Continuous variables were tested for normality with the Shapiro-Wilk test and using visual analysis of the distribution histograms. Those considered having a normal distribution were analyzed with the *t*-test, and nonparametric values were evaluated with the Wilcoxon Rank-Sum test. Categorical variables were analyzed using the chi-square test. Effect size for categorical variables was calculated with the odds ratio and presented with a 95% confidence interval (CI). The variable “days alive and free” was considered ordinal, and its effect size was calculated using ordered logistic regression with robust standard error. To compare the evolution of MAP over time between groups, we used a linear mixed-effects model with fixed effects for groups, time, and their interactions, and a random intercept for each patient. An autoregressive covariance structure of order 1 was selected based on the lowest Akaike information criterion, accounting for the temporal correlation of repeated measures. Estimation was performed using restricted maximum likelihood ([Supplementary topic 12](#)).

In contrast to the published protocol,¹² prespecified subgroup analyses were not performed due to the limited number of patients within each subgroup. However, a post hoc analysis was conducted in a nonprespecified subgroup of patients with a history of chronic hypertension (CH). Another post hoc analysis was performed to address the primary outcome according to a different adaptation to the Kidney Disease: Improving Global Outcomes criteria, considering both a 0.3 mg/dL absolute increase or a 1.5 $\times$ relative increase in serum creatinine levels. Missing data are described for each variable in the [Supplementary table 1](#) and were managed by simple exclusion. All statistical tests were executed with STATA, version 18 (Stata-Corp, College Station, USA). The significance level was 0.05. Tests and CIs were not adjusted for multiple comparisons; thus, except for the primary outcome, all analyses should be considered exploratory.

3. Results

3.1. Baseline data

Between August 27, 2021, and December 5, 2023, a total of 364 patients were screened following ICU admission after liver transplantation. After applying exclusion criteria, 174 patients were randomized and completed the trial, 89 in the higher MAP target group and 85 in the lower MAP target group ([Fig. 1](#)). The median age of the sample was 60 years, and 90.8% were

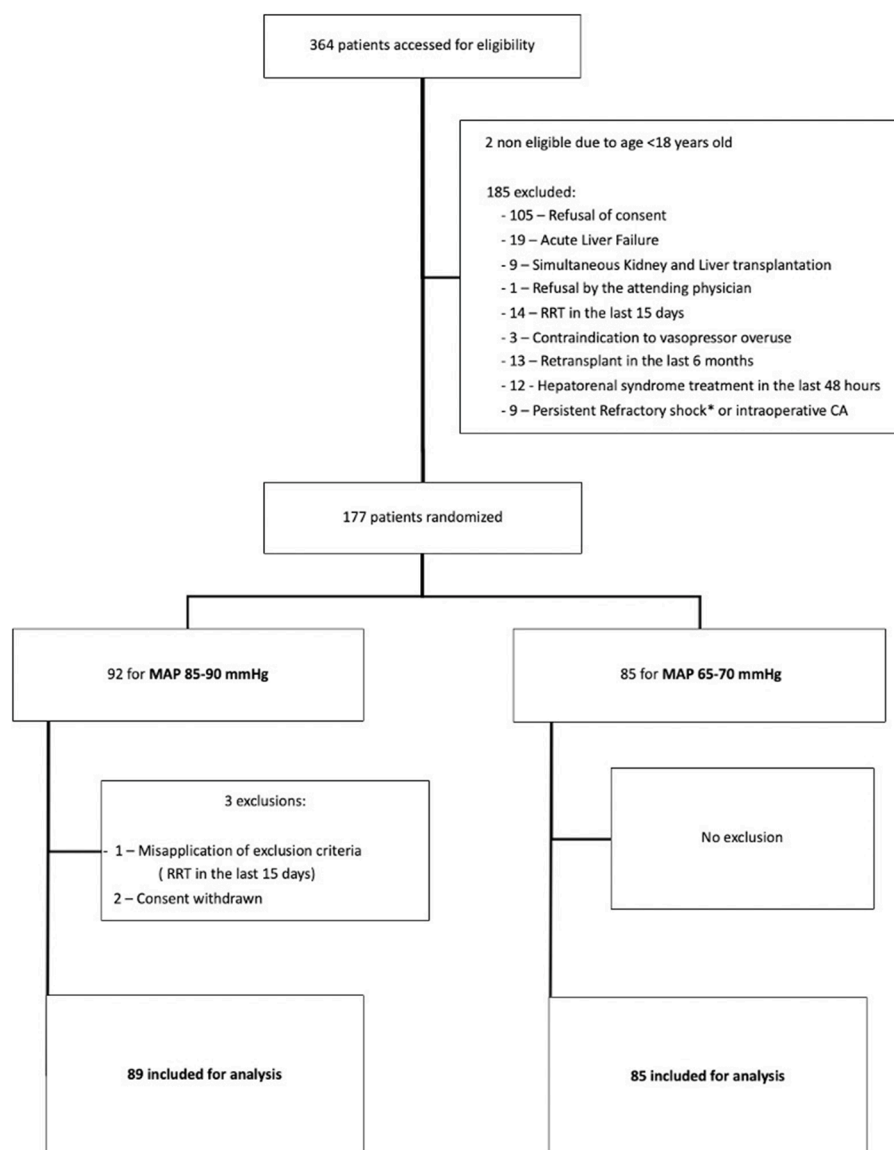


Figure 1. Research flowchart. MAP, mean arterial pressure; RRT, renal replacement therapy; CA, Cardiac Arrest.

outpatients. Regarding donors, 96% were donations after brainstem death, whereas 4% were living donors. Additional epidemiologic characteristics and intraoperative management details are provided in [Table 1](#).

3.2. Care processes and blood pressure

Data related to patient management during the intervention period are presented in [Table 2](#). The intervention group received a greater volume of fluids and had higher urinary output compared to the control group. Additionally, patients in the intervention group required higher maximum doses of norepinephrine and remained on vasopressor support for a longer duration during the first postoperative day ([Table 2](#)). Median MAP values and their distribution over the first 24 hours are illustrated in [Figure 2](#), along with corresponding 95% CIs. The mean MAP differed significantly between the groups, with a mean difference of 7 mmHg; 95% CI, -7.77 to -6.22 ; $P < .01$.

3.3. Primary outcome

The incidence of AKI within 7 days was similar between groups, occurring in 62 patients (69.7%) in the intervention group and 59 patients (69.4%) in the control group (relative risk [RR], 1.00; 95% CI, 0.82-1.22; $P = .97$). The daily distribution of AKI is presented in the [Supplementary figure 1](#), with the majority occurring on the first postoperative day (43.8% in the intervention group and 53% in the control group).

3.4. Secondary outcomes

Among the 13 secondary outcomes, the incidence of MAKE28 was lower in the intervention group (50 [56.1%] vs 62 [72.9%]; RR, 0.77; 95% CI, 0.61-0.96; $P = .02$). The incidence of each MAKE28 subcomponent is detailed in the [Supplementary figure 4](#). Notably, the occurrence of graft rejection requiring corticosteroid pulse therapy within 7 days was higher in the

Table 1

Basal data at ICU admission.

Parameter	MAP 65-70 mmHg N = 85	MAP 85-90 mmHg N = 89
Age (y)	57.1 (11.9)	57.2 (12.1)
Male sex, n (%)	59 (69.4)	50 (56.2)
SAPS III	46.2 (14.3)	47.2 (13.2)
SOFA	12 (10-13)	12 (9-14)
Charlson Comorbidity Index	5 (4-6)	5 (4-6)
Calculated MELD score	16 (11-22)	16 (11-20)
Listed MELD^a score	29 (24-29)	29 (25-29)
Refractory ascites, n (%)	14 (16.4)	25 (28)
Hepatocellular carcinoma, n (%)	40 (47.1)	33 (37.1)
BMI (kg/m²)	26 (23-30)	26 (23-29)
Outpatient, n (%)	77 (90.5)	81 (91.0)
Basal serum creatinine concentration (mg/dL)	0.85 (0.7-1)	0.86 (0.69-1.1)
Chronic hypertension, n (%)	33 (38.8)	29 (32.5)
Diabetes mellitus, n (%)	32 (37.6)	24 (26.9)
Cirrhosis main etiology		
Alcohol, n (%)	22 (25.9)	20 (22.5)
Hepatitis C, n (%)	27 (31.8)	19 (21.3)
MASH, n (%)	12 (14.1)	10 (11.2)
Cryptogenic, n (%)	7 (8.2)	13 (14.6)
Primary biliary cirrhosis, n (%)	1 (1.2)	7 (7.9)
Others, n (%)	16 (18.8)	20 (22.5)
Donor risk index	1.689 (1.338-1.939)	1.673 (1.382-1.899)
Donation after brainstem death, n (%)^b	79 (92.9)	87 (97.7)
Intraoperative data		
Maximum norepinephrine dosage (µg/kg/min)	0.25 (0.15-0.38)	0.3 (0.15-0.4)
Fluid balance ^c	2725 (1575-3930)	2500 (−270-3950)
Ischemia-reperfusion syndrome, n (%)	24 (28.9)	31 (35.2)
Maximum serum lactate concentration (mg/dL)	36 (29-47)	35 (28-46)
WIT (min)	34.9 (6.7)	34.2 (6.0)
CIT (h)	7.2 (6-8.3)	6.8 (5.8-7.9)

Table 1 (continued)

Parameter	MAP 65-70 mmHg N = 85	MAP 85-90 mmHg N = 89
Crystalloids (mL)	3500 (3000-4500)	3700 (3000-4500)
20% albumin (mL)	250 (150-400)	300 (200-450)
Urine output (mL)	1000 (600-1500)	1000 (600-1500)
Red blood cell units infused (unit)	1 (0-2)	2 (0-3)
Fibrinogen concentrate infusion (g)	0 (0-4)	0 (0-3)

Parametric continuous variables are presented as mean and (±standard deviation); nonparametric values are shown as median and (interquartile range, 25%-75%). Categorical variables are reported as frequency N and (%). Missing data were >5% for donor risk index, maximum norepinephrine dosage (µg/kg/min), urine output, and crystalloids ([Supplementary figure 3](#)).

BMI, body mass index; CIT, cold ischemia time; ICU, intensive care unit; MAP, mean arterial pressure; MASH, metabolic steatohepatitis; MELD, model for end-stage liver disease; mmHg, millimeters of mercury; SAPS III, simplified acute physiology score III; SOFA, sequential organ failure assessment; WIT, warm ischemia time.

^a According to current Brazilian legislation, additional points are added to the MELD score on the transplant waiting list for certain special conditions, such as hepatocellular carcinoma, refractory ascites, and refractory hepatic encephalopathy, among others.

^b All donors not classified as organ donation after brainstem death were living donors. Organ donation after circulatory death is not routinely performed in Brazil, and split grafts are not commonly used in our center.

^c Intraoperative fluid balance was calculated based on the sum of fluids and transfusions received minus the sum of diuresis and ascites. Data regarding intraoperative bleeding were not available.

intervention group (11 [12.5%] vs 3 [3.5%]; RR, 3.54; 95% CI, 1.02-12.25; $P = .03$).

The need for RRT, the number of days alive and free of AKI within 7 days, and the incidence of AKI assessed by measuring urinary neutrophil gelatinase-associated lipocalin levels at 48 hours were similar between groups ([Table 3](#)). The daily distribution of RRT requirements is provided in the [Supplementary table 3](#).

The incidence of EAD was 18% (16 patients) in the intervention group and 27.1% (23 patients) in the control group (RR, 0.65; 95% CI, 0.37-1.16; $P = .15$). PNF, HAT, and retransplantation were rare and similar between groups ([Table 3](#)). Additionally, 28-day survival, surgical site infection at 7 days, days alive and free from ICU at 7 days, and days alive and free from hospital at 28 days were also similar between groups.

Regarding safety outcomes, there were no significant differences between groups in the incidence of major bleeding or the need for blood product transfusion within 7 days. However, the

Table 2

Care during the first postoperative day and nephrotoxic exposure within the first week.

Parameter	MAP 65-70	MAP 85-90	P
	mmHg	mmHg	value
	N = 85	N = 89	
Crystalloids (mL)	1424.3 (763.18)	1658.5 (770.02)	.04
20% albumin (mL)	100 (0-100)	100 (0-100)	.70
Maximum norepinephrine dosage on 1POD (µg/kg/min)	0.18 (0.11-0.30)	0.26 (0.14-0.44)	<.01
Hours requiring vasopressors on 1POD	14 (7-24)	24 (15-24)	<.01
Abdominal drain output (mL)	330 (180-520)	300 (170-600)	.72
Urine output (mL)	500 (300-1040)	800 (440-1050)	.03
Fluid balance (mL)	602 (38-1500)	953 (392-1817)	.06
Basiliximab induction, n (%)	54 (63.5)	55 (61.7)	.81
Nephrotoxic exposure			
FK maximum level at 7 d (ng/mL)	7 (5-9.5)	6.8 (4.9-10)	.80
Other nephrotoxic ^a at 7 d, n (%)	27 (31.8)	34 (38.2)	.79

Parametric continuous variables are presented as mean ($\pm$ standard deviation); nonparametric values are shown as median and (interquartile range, 25%-75%). Categorical variables are reported as frequency N and (%).

Missing data were <5% in all variables.

FK, tacrolimus; MAP, mean arterial pressure; 1POD, first postoperative day.

^a Vancomycin, amikacin, acyclovir, polymyxin, or angiotensin converting enzyme inhibitors/angiotensin II receptor blocker.

incidence of arrhythmia requiring pharmacologic or electrical intervention was lower in the intervention group (2.2% vs 10.6%; RR, 0.21; 95% CI, 0.04-0.95; $P = .02$).

Renal outcomes were further explored in a post hoc analysis by stratifying patients based on the presence of CH (see [Table 3](#) and [supplementary table 4](#)). The effect size of the higher MAP strategy appeared more pronounced among patients with CH across several renal outcomes; however, none of these findings reached statistical significance compared with the original cohort. The incidence of AKI within 7 days, defined as either an absolute increase in serum creatinine levels of ≥ 0.3 mg/dL or a relative increase of ≥ 1.5 times, was also similar between groups ([supplementary table 4](#)).

4. Discussion

In this randomized, controlled, single-center trial, targeting a higher MAP (85-90 mmHg), compared with the conventional

target of 65 to 70 mmHg, did not reduce the incidence of AKI within 7 days following liver transplantation. Exploratory analysis of secondary renal outcomes revealed a lower incidence of MAKE28 in the intervention group. Additional outcomes were similar between groups, except for a higher rate of suspected graft rejection requiring pulse therapy within 7 days in the intervention group. Importantly, the higher MAP strategy was not associated with increased rates of major bleeding or transfusion requirements and was linked to a lower incidence of arrhythmias.

It is noteworthy that the median age of the study population was 60 years, and most patients were males (62%); the mean simplified acute physiology score III was 46.7 ([Table 1](#)), factors that may have contributed to the high observed incidence of AKI in this cohort.^{1,18} During the intervention period, crystalloid infusion and urinary output were modestly higher in the 85 to 90 mmHg group, findings compatible with the study intervention,¹⁹ as treating physicians were allowed to individualize fluid and/or vasopressor to achieve the assigned MAP target ([Table 2](#)).

Although the median MAP in the control group frequently exceeded the target (65-70 mmHg) ([Fig. 2](#)), it remained statistically lower than that of the intervention group ($P < .01$). This likely reflects a spontaneous rise in MAP even after complete vasopressor weaning. A similar MAP distribution was reported by previous trials,^{8,20} supporting the rationale for selecting a wider target difference in the present study, an approach that warrants consideration in future studies.

The primary outcome (AKI within 7 days) did not differ significantly between groups, suggesting a limited role for the intervention in preventing AKI. The post hoc analysis using a more sensitive criterion also showed similar outcomes between groups ([Supplementary figure 4](#)). The high incidence of AKI observed on postoperative day 1 (53% control, 43.8% intervention), also pointed out by previous data,²¹ may indicate that kidney injury predominantly occurred earlier, during the intraoperative phase, potentially limiting the effectiveness of a purely postoperative intervention. Implementing this strategy in the intraoperative period may have better highlighted its preventive potential, given the established association between intraoperative hypotension and postoperative AKI.²² Although judicious use of vasopressors during liver transplant is suggested to avoid bleeding, diastolic dysfunction, and graft ischemia,²³ the safety threshold remains uncertain. Notably, some studies have reported potential renal benefit from intraoperative administration of terlipressin, which appears to reduce portal venous flow without negatively impacting liver function tests.²⁴

Analysis of secondary outcomes revealed a lower incidence of MAKE28 in the intervention group, primarily driven by a reduction in persistent AKI (see [Supplementary table 3](#)). This finding suggests that targeting a higher MAP was not effective in preventing AKI within 7 days or improving early renal outcomes, but it may improve mid-term AKI. Although no previous MAP-targeting trial has been conducted specifically in liver transplant recipients, a higher MAP target (80-85 mmHg) in patients with cirrhosis with septic shock and AKI was associated with greater AKI reversal compared with usual care in a per-protocol

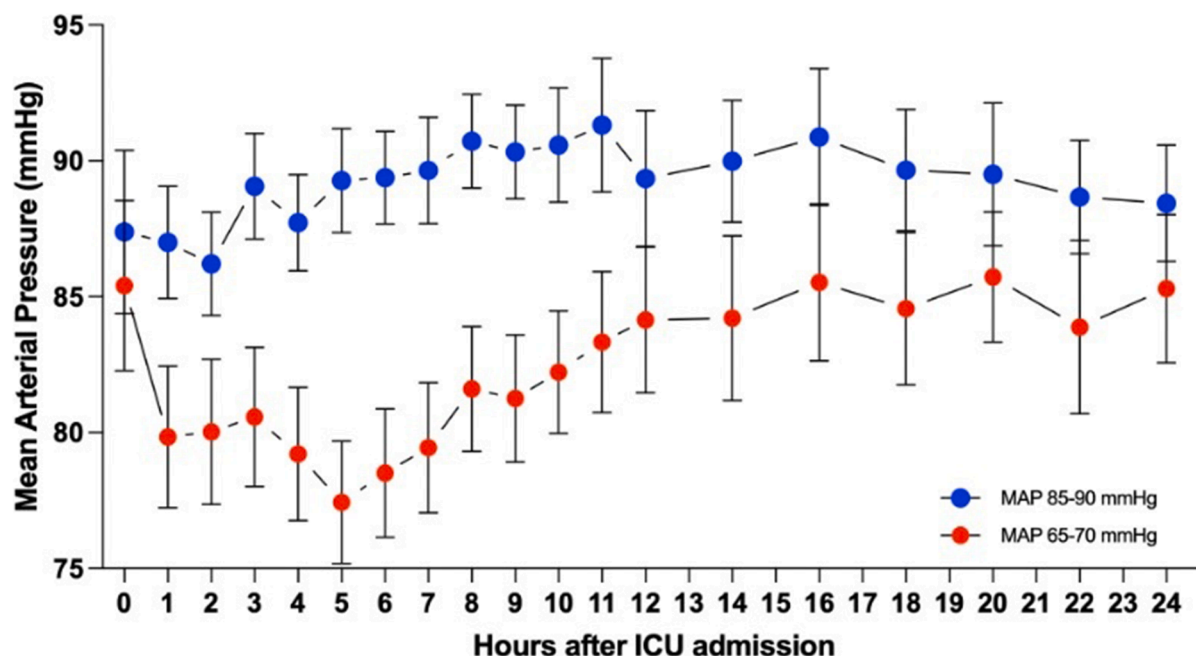


Figure 2. Blood pressure distribution between groups. ICU, intensive care unit; MAP, mean arterial pressure.

analysis, 53% vs 31% ($P = .01$).⁹ Similarly, a subanalysis of a prospective cohort in patients with decompensated cirrhosis demonstrated an association between higher MAP and AKI reversal.²⁵ These findings may be relevant to liver transplant recipients, as cirrhotic hemodynamic abnormalities often persist in the immediate postoperative period.²⁶ However, secondary outcomes should be interpreted with caution, as no adjustment for multiple comparisons was performed, increasing the risk that some observed differences may have occurred by chance.

Hypotension is common on the first postoperative day, affecting approximately 32% of patients after noncardiac surgery and increasing the risk of myocardial and kidney injury.^{5,27} Therapeutic strategies targeting higher MAP in the postoperative setting are rarely explored in critical care trials. One intraoperative trial demonstrated that maintaining MAP closer to preoperative baseline levels reduced postoperative complications, with a nonsignificant trend toward lower AKI incidence (10.9% vs 17.9%, $P = .09$).²⁸ Similarly, a nephroprotective effect of a higher intraoperative MAP (80–95 mmHg) was observed in older patients undergoing abdominal surgery compared with both lower and excessively high targets, suggesting a potential U-shaped relationship.¹⁴ However, these findings remain controversial and lack consensus. A trial comparing intraoperative MAP targets >60 vs >75 mmHg during noncardiac surgery found that, although intraoperative hypotension was associated with adverse outcomes, aiming for higher MAP targets did not reduce cardiovascular and renal complications.²⁹ However, in older patients with septic shock, a trial targeting higher MAP was associated with a significant increase in mortality compared with standard care.²⁰

The post hoc analysis of renal outcomes in the subgroup of patients with CH (Table 3 and Supplementary Material)

suggested a potentially greater nephroprotective effect of the higher MAP strategy. Although exploratory, this finding is consistent with the results reported by Pedro et al⁸ and may help explain the benefits observed by Futier et al,²⁸ in which 82.2% of participants had CH.

Although liver function was not commonly evaluated in previous MAP-targeting trials, our findings support the safety of this strategy. Despite requiring higher doses and longer durations of norepinephrine on the first postoperative day (Table 2), the intervention group did not show significant differences in rates of primary graft dysfunction (EAD or PNF), retransplantation, or HAT. However, a higher incidence of corticosteroid pulse therapy due to clinical suspicion of cellular rejection was observed in the interventional group (12.5% vs 3.5%; RR, 3.54; 95% CI, 1.02–12.25; $P = .030$). This finding may represent a spurious association due to multiple unadjusted comparisons. Alternatively, it is biologically plausible that ischemia of the bile ducts, a territory not protected by the liver's dual blood supply³⁰ and sensitive to norepinephrine,³¹ could lead to elevations in canalicular enzymes. This biochemical pattern may be misinterpreted as cellular rejection, as canalicular enzyme elevation is a key diagnostic marker for graft rejection.^{31,32} Notably, a previous cohort study found no association between vasopressor use and biliary complications following liver transplantation.³³

Survival, incidence of infection, and ICU/hospital-free days were similar between groups. It is important to note that the study was not powered to detect differences in these secondary outcomes. This information is consistent with previous studies, which have also reported no significant impact of this strategy on mortality or length of stay.^{8,9,29}

The selection of safety outcomes was guided by the potential risks associated with higher MAP values on bleeding³⁴ and

Table 3

Outcomes.

Parameter	MAP 65-70 mmHg N = 85	MAP 85-90 mmHg N = 89	Odds ratio (95% CI)	P value
Primary outcome				
AKI within 7 d, n (%)	59 (69.4)	62 (69.7)	1.00 (0.82-1.22)	.97
Renal secondary outcome				
MAKE28, n (%)	62 (72.9)	50 (56.1)	0.77 (0.61-0.96)	.02
Chronic hypertension, n (%) ^a	26/33 (78.7)	12/29 (41.3)	0.19 (0.05-0.65)	<.01
No chronic hypertension, n (%) ^a	36/52 (69.2)	38/60 (63.3)	0.76 (0.32-1.81)	0.51
Renal replacement therapy at 7 d, n (%)	16 (18.8)	11 (12.4)	0.65 (0.32-1.33)	.23
Chronic hypertension, n (%) ^a	8/33 (24.2)	2/29 (6.8)	0.23 (0.02-1.34)	.06
No chronic hypertension, n (%) ^a	8/52 (15.3)	9/60 (15)	0.97 (0.30-3.16)	.95
Days Alive and Free of AKI at 7 d ^b	4 (0-7)	5 (1-7)	1.36 (0.79-2.32)	.25
uNGAL >150 pg/mL at 48 h, n (%) ^c	20 (28.1)	16 (22.2)	0.78 (0.44-1.39)	.41
Hepatic secondary outcomes, n (%)				
Primary dysfunction, EAD	23 (27.1)	16 (18)	0.66 (0.37-1.16)	.15
Primary dysfunction, PNF	1 (1.2)	2 (2.2)	1.91 (0.17-20.67)	.58
Hepatic artery thrombosis at 28 d	2 (2.4)	3 (3.4)	1.43 (0.24-8.36)	.68

Table 3 (continued)

Parameter	MAP 65-70 mmHg N = 85	MAP 85-90 mmHg N = 89	Odds ratio (95% CI)	P value
Graft rejection requiring pulse therapy at 7 d	3 (3.5)	11 (12.5)	3.54 (1.02-12.25)	.03
Retransplant at 28 d	2 (2.4)	3 (3.4)	1.43 (0.24-8.36)	.68
Clinical secondary outcomes				
Survival - 28 d, n (%)	77 (90.6)	78 (87.6)	1.31 (0.55-3.10)	.53
Days alive and free of ICU at 7 d ^b	1.9 (0-3.3)	2.3 (0-3.6)	1.29 (0.76-2.21)	.33
Days alive and free of hospital at 28 d ^b	16 (1-19)	14 (0-19)	0.96 (0.57-1.61)	.87
Surgical site infection at 7 d, n (%)	3 (3.5)	7 (7.9)	2.22 (0.59-8.33)	.21
Safety outcomes, n (%)				
Major bleeding at 7 d	6 (7.1)	8 (9)	1.27 (0.46-3.51)	.63
Blood product requirement at 7 d	0 (0-1)	0 (0-1)	0.89 (0.49-1.62)	.71
Arrhythmia requiring intervention at 7 d	9 (10.6)	2 (2.2)	0.21 (0.04-0.95)	.02

Parametric continuous variables are presented as mean ($\pm$ standard deviation); nonparametric values are shown as median and (interquartile range, 25%-75%). Categorical variables are reported as frequency N (%).

Missing data were >5% for uNGAL level measurements at 48 hours ([Supplementary Material](#)).

AKI, acute kidney injury; ICU, intensive care unit; EAD, early allograft dysfunction; MAKE28, major adverse kidney events at 28 days; PNF, primary nongraft dysfunction; RRT, renal replacement therapy; uNGAL, urinary neutrophil gelatinase-associated lipocalin.

^a Post hoc analysis. The complete description of all renal outcomes in relation to previous chronic hypertension can be found in the [Supplementary Material](#).

^b After surgery.

^c Patients on renal replacement therapy or dead were excluded from analysis.

arrhythmias.⁸ The intervention was found to be safe with no significant difference in the incidence of major bleeding or transfusion requirements during the first week, findings consistent with previous trials on abdominal surgery and septic shock.^{8,28} The incidence of arrhythmias requiring pharmacologic or electrical intervention within 7 days was lower in the higher MAP group. This result contrasts with findings reported by Asfar et al⁸ but is in line with a trial evaluating early norepinephrine initiation in septic shock patients,³⁵ suggesting a possible arrhythmogenic role of hypotension.³⁶

This study has several limitations; its single-center design is insufficient to definitively answer the research question. Additionally, blinding the groups was not feasible and may induce bias. The pragmatic fluid management protocol and the lack of data regarding cardiac output, pulmonary pressure, and intraoperative MAP may limit findings interpretation. Restricting the intervention to the postoperative period may have missed a critical window for AKI prevention. The MAP in the control groups was higher than planned, as the authors considered it unsafe to use vasodilators when MAPs were spontaneously above the target values. Although our approach was also adopted in previous trials,^{8,9,28} it may have limited the ability to maintain strict MAP separation between groups, potentially reducing the study's power to detect significant differences. The incidence of the primary outcome was slightly lower than anticipated (69% vs 75%), although the similarity in AKI rates between groups challenges the possibility of the study being underpowered. On the other hand, the incidence of AKI within 7 days was high (69%), which may limit the generalizability of the findings. Reported rates of posttransplant AKI have varied widely, ranging from 15% to 81%. This variability likely reflects differences in the diagnostic criteria used over the past decades—with more recent definitions being more sensitive—as well as variations in the postoperative timeframes considered (eg, 24 hours vs 7 days) and specific institutional or sample characteristics. The study's primary outcome definition for AKI deviated from both Kidney Disease: Improving Global Outcomes³⁷ and cirrhotic³⁸ criteria, as oliguria is often considered unreliable in the surgical context,^{39,40} and might be unreliable in cases with high abdominal drain debts. Additionally, pulse therapy was prescribed by an external member from the surgical team based on case-by-case analysis without a predefined cutoff for trial participants. Finally, the absence of adjustments for multiple comparisons limits the interpretability of secondary outcomes.

Despite its limitations, this study represents a rare randomized controlled trial conducted in the postoperative setting of liver transplantation. It evaluates a simple, pragmatic, and feasible hemodynamic strategy against a common and serious complication.

In conclusion, targeting a higher MAP (85–90 mmHg) during the first 24 hours after liver transplantation did not reduce the incidence of AKI within 7 days compared with the standard MAP target (65–70 mmHg). Exploratory secondary outcomes indicate a possible association between higher MAP and reduced MAKE28, which may merit further investigation in future studies.

Funding

This research did not receive any specific grant or funding. None of the authors received any financial support to conduct this trial.

Declaration of competing interest

The authors of this manuscript have no conflicts of interest to disclose as described by *American Journal of Transplantation*.

Data availability

The data from this trial is accessible upon institutional approval and a reasonable request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used ChatGPT-4.0 exclusively in order to assist with language and readability improvements, as none of the authors are native English speakers. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ajt.2025.07.2482>.

ORCID

Rodolpho Augusto de Moura Pedro  <https://orcid.org/0000-0003-2013-6151>
 Bruna Carla Scharanch  <https://orcid.org/0009-0009-9674-4592>
 Lucas de Oliveira Araújo  <https://orcid.org/0000-0001-8073-9443>
 Paula Sepulveda Mesquita  <https://orcid.org/0000-0001-9749-370X>
 Luciana Severo Brandão  <https://orcid.org/0009-0006-8859-3067>
 Lúcia da Conceição Andrade  <https://orcid.org/0000-0002-3078-8957>
 Wellington Andraus  <https://orcid.org/0000-0002-5162-138X>
 Luís Augusto Carneiro D'Albuquerque  <https://orcid.org/0000-0001-7607-7168>
 Jean-Michel Constantin  <https://orcid.org/0000-0001-7634-5898>
 Alexandre Biasi Cavalcanti  <https://orcid.org/0000-0003-2798-6263>
 Alberto Queiroz Faria  <https://orcid.org/0000-0002-5572-663X>
 Luiz Marcelo Sá Malbouisson  <https://orcid.org/0000-0002-3261-5603>

References

1. Thongprayoon C, Kaewput W, Thamcharoen N, et al. Incidence and impact of acute kidney injury after liver transplantation: a meta-analysis. *J Clin Med*. 2019;8(3):372. <https://doi.org/10.3390/jcm8030372>.
2. Barreto AG, Daher EF, Silva Junior GB, et al. Risk factors for acute kidney injury and 30-day mortality after liver transplantation. *Ann Hepatol*. 2015; 14(5):688–694. [https://doi.org/10.1016/S1665-2681\(19\)30763-X](https://doi.org/10.1016/S1665-2681(19)30763-X).
3. Dong V, Nadim MK, Karvellas CJ. Post-liver transplant acute kidney injury. *Liver Transpl*. 2021;27(11):1653–1664. <https://doi.org/10.1002/lt.26094>.
4. Zhou J, Zhang X, Lyu L, Ma X, Miao G, Chu H. Modifiable risk factors of acute kidney injury after liver transplantation: a systematic review and

- meta-analysis. *BMC Nephrol.* 2021;22(1):149. <https://doi.org/10.1186/s12882-021-02360-8>.
5. Walsh M, Devereaux PJ, Garg AX, et al. Relationship between intraoperative mean arterial pressure and clinical outcomes after noncardiac surgery: toward an empirical definition of hypotension. *Anesthesiology.* 2013;119(3):507–515. <https://doi.org/10.1097/ALN.0b013e3182a10e26>.
 6. Mohmand H, Goldfarb S. Renal dysfunction associated with intra-abdominal hypertension and the abdominal compartment syndrome. *J Am Soc Nephrol.* 2011;22(4):615–621. <https://doi.org/10.1681/ASN.2010121222>.
 7. Varpula M, Tallgren M, Saukkonen K, Voipio-Pulkki LM, Pettilä V. Hemodynamic variables related to outcome in septic shock. *Intensive Care Med.* 2005;31(8):1066–1071. <https://doi.org/10.1007/s00134-005-2688-z>.
 8. Asfar P, Meziani F, Hamel JF, et al. High versus low blood-pressure target in patients with septic shock. *N Engl J Med.* 2014;370(17):1583–1593. <https://doi.org/10.1056/NEJMoa1312173>.
 9. Maiwall R, Rao Pasupuleti SS, Hidam AK, et al. A randomised-controlled trial (TARGET-C) of high vs. low target mean arterial pressure in patients with cirrhosis and septic shock. *J Hepatol.* 2023;79(2):349–361. <https://doi.org/10.1016/j.jhep.2023.04.006>.
 10. Skytte Larsson J, Bragadottir G, Redfors B, Ricksten SE. Renal effects of norepinephrine-induced variations in mean arterial pressure after liver transplantation: a randomized cross-over trial. *Acta Anaesthesiol Scand.* 2018;62(9):1229–1236. <https://doi.org/10.1111/aas.13156>.
 11. Kwon HM, Moon YJ, Jung KW, et al. Low mean arterial blood pressure is independently associated with postoperative acute kidney injury after living donor liver transplantation: a propensity score weighing analysis. *Ann Transplant.* 2018;23:236–245. <https://doi.org/10.12659/AOT.908329>.
 12. Pedro RAM, Scharranch BC, Araújo LO, et al. High versus low mean arterial pressure target in liver transplant patients. An open, controlled, single-center, randomized clinical trial - protocol and methods (LIVER-PAM). *Crit Care Sci.* 2023;35(1):11–18. <https://doi.org/10.5935/2965-2774.20230336-en>.
 13. Cobos-Carbó A, Augustovski F. [CONSORT 2010 declaration: updated guideline for reporting parallel group randomised trials]. *Med Clin (Barc).* 2011;137(5):213–215. <https://doi.org/10.1016/j.medcli.2010.09.034>.
 14. Wu X, Jiang Z, Ying J, Han Y, Chen Z. Optimal blood pressure decreases acute kidney injury after gastrointestinal surgery in elderly hypertensive patients: a randomized study: optimal blood pressure reduces acute kidney injury. *J Clin Anesth.* 2017;43:77–83. <https://doi.org/10.1016/j.jclinane.2017.09.004>.
 15. See EJ, Toussaint ND, Bailey M, et al. Risk factors for major adverse kidney events in the first year after acute kidney injury. *Clin Kidney J.* 2021;14(2):556–563. <https://doi.org/10.1093/ckj/sfz169>.
 16. Aklilu AM, Kumar S, Nugent J, et al. COVID-19-associated acute kidney injury and longitudinal kidney outcomes. *JAMA Intern Med.* 2024;184(4):414–423. <https://doi.org/10.1001/jamainternmed.2023.8225>.
 17. Fuhrman DY, Stenson EK, Alhamoud I, et al. Major adverse kidney events in pediatric continuous kidney replacement therapy. *JAMA Netw Open.* 2024;7(2):e240243. <https://doi.org/10.1001/jamanetworkopen.2024.0243>.
 18. Park MH, Shim HS, Kim WH, et al. Clinical risk scoring models for prediction of acute kidney injury after living donor liver transplantation: a retrospective observational study. *PLoS One.* 2015;10(8):e0136230. <https://doi.org/10.1371/journal.pone.0136230>.
 19. Myles PS, McLroy DR, Bellomo R, Wallace S. Importance of intraoperative oliguria during major abdominal surgery: findings of the restrictive versus liberal fluid therapy in major abdominal surgery trial. *Br J Anaesth.* 2019;122(6):726–733. <https://doi.org/10.1016/j.bja.2019.01.010>.
 20. Endo A, Yamakawa K, Tagami T, et al. Efficacy of targeting high mean arterial pressure for older patients with septic shock (OPTPRESS): a multicentre, pragmatic, open-label, randomised controlled trial. *Intensive Care Med.* 2025;51(5):883–892. <https://doi.org/10.1007/s00134-025-07910-4>.
 21. Rahman S, Davidson BR, Mallett SV. Early acute kidney injury after liver transplantation: predisposing factors and clinical implications. *World J Hepatol.* 2017;9(18):823–832. <https://doi.org/10.4254/wjh.v9.i18.823>.
 22. Joosten A, Lucidi V, Ickx B, et al. Intraoperative hypotension during liver transplant surgery is associated with postoperative acute kidney injury: a historical cohort study. *BMC Anesthesiol.* 2021;21(1):12. <https://doi.org/10.1186/s12871-020-01228-y>.
 23. Kramer DJ, Siegal EM, Frogge SJ, Chadha MS. Perioperative management of the liver transplant recipient. *Crit Care Clin.* 2019;35(1):95–105. <https://doi.org/10.1016/j.ccc.2018.08.012>.
 24. Fayed N, Refaat EK, Yassein TE, Alwaraqy M. Effect of perioperative terlipressin infusion on systemic, hepatic, and renal hemodynamics during living donor liver transplantation. *J Crit Care.* 2013;28(5):775–782. <https://doi.org/10.1016/j.jcrc.2013.02.016>.
 25. Cullaro G, Allegretti AS, Fenton C, et al. The association between mean arterial pressure and acute kidney injury reversal among patients with decompensated cirrhosis. *Hepatology.* 2025;81(1):126–135. <https://doi.org/10.1097/HEP.0000000000000858>.
 26. Al-Hamoudi WK, Alqahtani S, Tandon P, Ma M, Lee SS. Hemodynamics in the immediate post-transplantation period in alcoholic and viral cirrhosis. *World J Gastroenterol.* 2010;16(5):608–612. <https://doi.org/10.3748/wjg.v16.i5.608>.
 27. Sessler DI, Meyhoff CS, Zimmerman NM, et al. Period-dependent associations between hypotension during and for four days after noncardiac surgery and a composite of myocardial infarction and death: a substudy of the POISE-2 trial. *Anesthesiology.* 2018;128(2):317–327. <https://doi.org/10.1097/ALN.0000000000001985>.
 28. Futier E, Lefrant JY, Guinot PG, et al. Effect of individualized vs standard blood pressure management strategies on postoperative organ dysfunction among high-risk patients undergoing major surgery: a randomized clinical trial. *JAMA.* 2017;318(14):1346–1357. <https://doi.org/10.1001/jama.2017.14172>.
 29. Wanner PM, Wulff DU, Djurdjevic M, Korte W, Schnider TW, Filipovic M. Targeting higher intraoperative blood pressures does not reduce adverse cardiovascular events following noncardiac surgery. *J Am Coll Cardiol.* 2021;78(18):1753–1764. <https://doi.org/10.1016/j.jacc.2021.08.048>.
 30. Ramesh Babu CS, Sharma M. Biliary tract anatomy and its relationship with venous drainage. *J Clin Exp Hepatol.* 2014;4(Suppl 1):S18–S26. <https://doi.org/10.1016/j.jceh.2013.05.002>.
 31. van Limmen J, Iturriagagoitia X, Verougstraete M, et al. Effect of norepinephrine infusion on hepatic blood flow and its interaction with somatostatin: an observational cohort study. *BMC Anesthesiol.* 2022;22(1):202. <https://doi.org/10.1186/s12871-022-01741-2>.
 32. Rodríguez-Perálvarez M, Germani G, Tsochatzis E, et al. Predicting severity and clinical course of acute rejection after liver transplantation using blood eosinophil count. *Transpl Int.* 2012;25(5):555–563. <https://doi.org/10.1111/j.1432-2277.2012.01457.x>.
 33. Larivière J, Giard JM, Zuo RM, Massicotte L, Chassé M, Carrier FM. Association between intraoperative fluid balance, vasopressors and graft complications in liver transplantation: a cohort study. *PLoS One.* 2021;16(7):e0254455. <https://doi.org/10.1371/journal.pone.0254455>.
 34. Kudo D, Yoshida Y, Kushimoto S. Permissive hypotension/hypotensive resuscitation and restricted/controlled resuscitation in patients with severe trauma. *J Intensive Care.* 2017;5(1):11. <https://doi.org/10.1186/s40560-016-0202-z>.
 35. Pempikul C, Tongyoo S, Viarasilpa T, Trainarongsakul T, Chakorn T, Udompanturak S. Early use of norepinephrine in septic shock resuscitation (CENSER). A randomized trial. *Am J Respir Crit Care Med.* 2019;199(9):1097–1105. <https://doi.org/10.1164/rccm.201806-1034OC>.
 36. Mc Causland FR, Tumlin JA, Roy-Chaudhury P, et al. Intradialytic hypotension and cardiac arrhythmias in patients undergoing maintenance hemodialysis: results from the monitoring in dialysis study. *Clin J Am Soc Nephrol.* 2020;15(6):805–812. <https://doi.org/10.2215/CJN.06810619>.

37. Khwaja A. KDIGO clinical practice guidelines for acute kidney injury. *Nephron Clin Pract.* 2012;120(4):c179–c184. <https://doi.org/10.1159/000339789>.
38. Nadim MK, Kellum JA, Forni L, et al. Acute kidney injury in patients with cirrhosis: acute disease quality initiative (ADQI) and International Club of Ascites (ICA) joint multidisciplinary consensus meeting. *J Hepatol.* 2024;81(1):163–183. <https://doi.org/10.1016/j.jhep.2024.03.031>.
39. Prowle JR, Forni LG, Bell M, et al. Postoperative acute kidney injury in adult non-cardiac surgery: joint consensus report of the acute disease quality initiative and perioperative quality initiative. *Nat Rev Nephrol.* 2021;17(9):605–618. <https://doi.org/10.1038/s41581-021-00418-2>.
40. de Moura Pedro RA, Cunha GB, Pietrobon I, et al. Should diuresis remain a diagnostic criterion for surgery associated acute kidney injury? Revisiting the role of perioperative oliguria. *J Crit Care.* 2025;87: 155047. <https://doi.org/10.1016/j.jcrc.2025.155047>.